

99.500 Applied Mathematics for Engineering

Spring Semester AY2026

Modeling & Nondimensionalization

Week 2 Outline

- Flow Reactor
- Axial Dispersion in Flow Reactor
- Batch Reactor
- Heat Transfer Semi-infinite block

For the flow reactor problem, we derived and discussed last week, how well do you understand it? Please rate your understanding on a scale of 1 to 5, where 1 means 'not understood at all' and 5 means 'fully understood'.

- A. 1
- B. 2
- C. 3
- D. 4
- E. 5



Multiple Choice

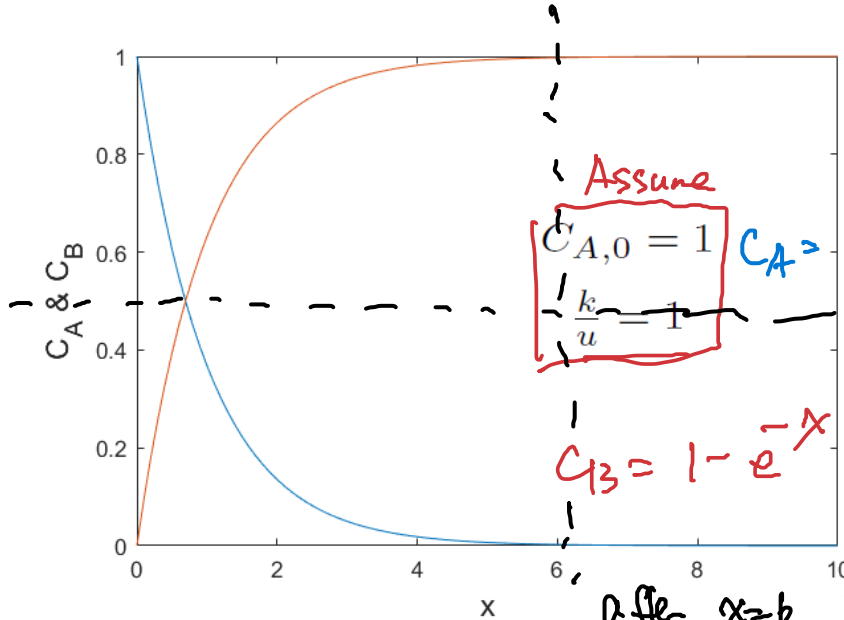
Flow Reactor

Separation of variables

Governing equation

$\frac{dC_A}{dx} = -\frac{k}{u} C_A$ (1st order rxn rate const) $\Rightarrow \int \frac{dC_A}{C_A} = \int -\frac{k}{u} dx$
 $C_A(x=0) = C_{A,0}$ (flow rate) $\Rightarrow \ln C_A = -\frac{k}{u} x + c_1 \Rightarrow C_A = c_2 e^{-\frac{k}{u} x}$ (initial condn at $x=0$)
 $C_A(x=0) = C_{A,0} \Rightarrow c_2 = C_{A,0}$
 $C_A = C_{A,0} e^{-\frac{k}{u} x}$ & $C_B = C_{A,0} (1 - e^{-\frac{k}{u} x})$

Boundary condition



$C_A + C_B = \text{const} = C_{A,0}$
 $C_B = C_{A,0} - C_{A,0} e^{-\frac{k}{u} x}$

What insights can we derive from the graph?

after $x=b$, rxn completed, later parts not useful.

What is Nondimensionalization?

A mathematical technique[?] used to remove physical units from equations by introducing dimensionless variables.
No units

- Simplifies equations *reduce complexity & makes numerical simulation more efficient.*
- Identifies key parameters *gives more physical meaning. recognize dominant effects*
- Generalizes solutions
- Improves numerical stability

Steps of Nondimensionalization

1. Identify characteristic scales for variables
2. Define dimensionless variables using these characteristic scales
3. Rewrite the original equation in terms of these new variables
4. Identify key dimensionless groups, which determine the system's behaviour

Flow Reactor : Nondimensionalization

$$\frac{dC_A}{dx} = -\frac{k}{u} C_A$$

$$C_A = \tilde{C}_A C_{A,0} \quad \text{scale for A's concentration}$$

$$\frac{d(\tilde{C}_A C_{A,0})}{d(\tilde{x} L)} = -\frac{k}{u} \tilde{C}_A C_{A,0}$$

$$\text{Let } \tilde{C}_A = \frac{C_A}{C_{A,0}} \text{ \& } \tilde{x} = \frac{x}{L} \rightarrow x = \tilde{x} L$$

The governing and BC equation is its dimensionless form:

$$\frac{d\tilde{C}_A}{d\tilde{x}} = -\underbrace{\frac{kL}{u}}_{\alpha} \tilde{C}_A \Rightarrow \frac{d\tilde{C}_A}{d\tilde{x}} = -\alpha \tilde{C}_A \text{ where } \alpha = \frac{kL}{u} \cdot \frac{\frac{1}{s} \cdot m}{\frac{m}{s}} = 1$$

$$\tilde{C}_A(\tilde{x} = 0) = 1 \text{ B.C.} \quad x=0 \rightarrow \tilde{x}=0$$

$$\alpha = \frac{kL}{u} = \frac{\text{rxn rate} \cdot \text{vol}}{\text{flow rate}} = \frac{\frac{\text{mol}}{\text{m}^3 \cdot \text{s}} \cdot \text{m}^3}{\frac{\text{mol}}{\text{s}}} = \frac{\text{mol}}{\text{s}} \cdot \frac{\text{m}^3}{\text{m}^3} = \frac{\text{mol}}{\text{s}}$$

$$C_A(x=0) = C_{A,0} \Rightarrow \tilde{C}_A(\tilde{x}=0) = 1$$

$\sim$ number of moles of A consumed in the reactor tube per unit time
 $\sim$ number of moles of A flowing through the reactor tube per unit time
 $\sim$ reaction rate for reactant A
 $\sim$ flow rate for reactant A

Flow Reactor: Result and Interpretation

Solve the dimensionless ODE:

$$\frac{d\tilde{C}_A}{d\tilde{x}} = -\alpha\tilde{C}_A \Rightarrow \int \frac{d\tilde{C}_A}{\tilde{C}_A} = \int -\alpha d\tilde{x}$$

1. 11
 $\tilde{x}=0, \tilde{C}_A = C_2 \cdot e^0 = C_2$
 How about B?

$$\frac{C_A + C_B}{C_{A,0}} = \frac{C_{A,0}}{C_{A,0}}$$

$$\ln \tilde{C}_A = -\alpha\tilde{x} + c_1 \Rightarrow \tilde{C}_A = c_2 e^{-\alpha\tilde{x}}$$

$$\tilde{C}_A(\tilde{x}=0) = 1 \Rightarrow c_2 = 1 \Rightarrow \tilde{C}_A = e^{-\alpha\tilde{x}}$$

$$\tilde{C}_A + \left[\frac{C_B}{C_{A,0}} \right] = 1$$

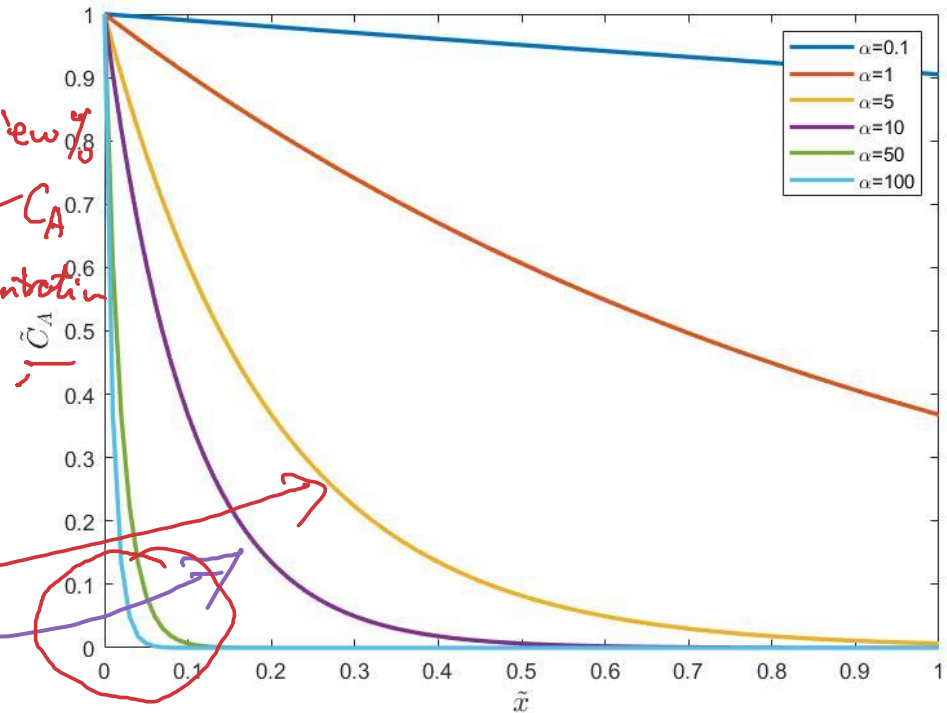
What insights can we derive from the graph?

view of C_A concentration

$$\alpha = \frac{kL}{u} = \frac{kC_{A,0}LA}{uC_{A,0}A}$$

if $L=6, \frac{k}{u}=1, \alpha=6$

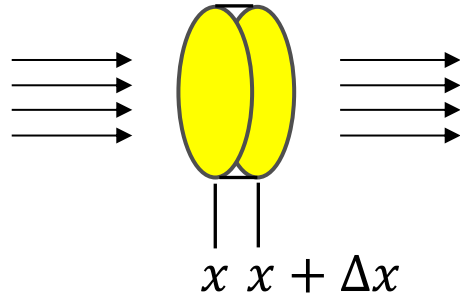
if $L=3, \alpha=10, \frac{k}{u}=\frac{10}{3}$



Axial Dispersion (AD)

Flow reactor with invalid assumption

due to conc. difference.
Dispersion term: proportional to concentration gradient



In:

$$uC_A|_x A - D \left. \frac{dC_A}{dx} \right|_x A$$

Out:

$$uC_A|_{x+\Delta x} A - D \left. \frac{dC_A}{dx} \right|_{x+\Delta x} A$$

Generation: 0

Consumption: $-r_A \Delta V = k C_A A \Delta x$

Accumulation: 0 for steady-state

Shell balance equation:

$$0 = \left[uC_A|_x A - D \left. \frac{dC_A}{dx} \right|_x A \right] - \left[uC_A|_{x+\Delta x} A - D \left. \frac{dC_A}{dx} \right|_{x+\Delta x} A \right] - k C_A A \Delta x$$

With re-arrangement:

$$D \frac{d^2 C_A}{dx^2} - u \frac{dC_A}{dx} - k C_A = 0$$

Boundary conditions:

$$uC_{A,0} = \left[uC_A - D \left. \frac{dC_A}{dx} \right| \right]_{x=0} \quad \& \quad \left[\frac{dC_A}{dx} \right]_{x=L} = 0 \quad (\text{Danckwerts BC})$$

diffusion $\rightarrow$ passive factor driven by conc gradient.

Combine, take limit $\Delta x \rightarrow 0 \Rightarrow D \frac{d^2 C_A}{dx^2}$

diffusion effects are negligible mean conc gradient at the outlet is 0.

AD: Nondimensionalization

$$x \rightarrow \tilde{x} \Rightarrow \tilde{x} \rightarrow 0$$

Let $\tilde{C}_A = \frac{C_A}{C_{A,0}}$ & $\tilde{x} = \frac{x}{L}$

$\alpha = \frac{D}{uL}$ $\beta = \frac{Lk}{u}$

$\frac{D}{uL} \frac{d^2 \tilde{C}_A}{d\tilde{x}^2} - \frac{D}{uL} \frac{d\tilde{C}_A}{d\tilde{x}} - \beta \tilde{C}_A = 0$

$\left[\tilde{C}_A - \alpha \frac{d\tilde{C}_A}{d\tilde{x}} \right]_{\tilde{x}=0} = 1$

$\left[\frac{d\tilde{C}_A}{d\tilde{x}} \right]_{\tilde{x}=1} = 0$

$x=L \Rightarrow \tilde{x} = \frac{L}{L} = 1$

$\frac{dC_A}{dx} = \frac{d(\tilde{C}_A \cdot C_{A,0})}{d(\tilde{x}L)} = 0$

$\frac{d\tilde{C}_A}{d\tilde{x}} \Big|_{\tilde{x}=1} = 0$

AD: Nondimensionalization

$\frac{dC_A}{dx} \rightarrow D \cdot \frac{dC_A}{dx} \cdot A \Rightarrow$ diffusi term.

$\alpha = \frac{D}{uL} = \frac{D (C_{A,0}/L) A}{u C_{A,0} A} \sim \frac{\text{AD diffusn.}}{\text{flow rate for reactant A}}$

When AD is negligible,

$\alpha \ll 1$

$\Rightarrow \begin{cases} -\tilde{C}'_A - \beta \tilde{C}_A = 0 \\ \tilde{C}_A(0) = 1 \end{cases} \quad \tilde{C}'_A(L) = 0$

When AD is not negligible (well mixing)

$\alpha \gg 1$

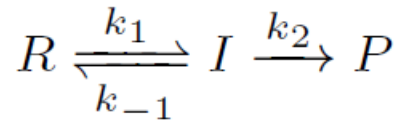
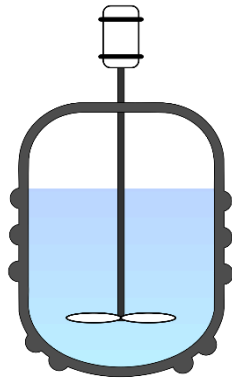
$\Rightarrow \begin{cases} \tilde{C}''_A = 0 \\ -\alpha \tilde{C}'_A(0) = 1 \end{cases} \quad \tilde{C}'_A(L) = 0$

at $\tilde{x}=0, \tilde{C}_A=1$

$\Rightarrow \tilde{C}_A \text{ is constant}$
 $\tilde{C}_A' = C_1 \rightarrow C_1=0$
 $\tilde{C}_A = C_2$

Reaction in Batch Reactor

Main assumptions: (1) a well-mixed solution; (2) Constant T; (3) The fluid is incompressible



rate of R.

↑

$$r_R = -k_1 C_R + k_{-1} C_I$$

→ R conc. → I conc.

$$r_I = k_1 C_R - k_{-1} C_I - k_2 C_I$$

↓

$$r_P = k_2 C_I$$

rate of product.

$$\begin{cases} \frac{dC_R}{dt} = -k_1 C_R + k_{-1} C_I \\ \frac{dC_I}{dt} = k_1 C_R - k_{-1} C_I - k_2 C_I \\ \frac{dC_P}{dt} = k_2 C_I \end{cases} \text{ with initial conditions } \begin{cases} C_R(0) = C_{R,0} \\ C_I(0) = 0 \\ C_P(0) = 0 \end{cases}$$

Reaction in Batch Reactor: Nondimensionalization

$$\text{Let } \tilde{C}_* = \frac{C_*}{C_{R,0}} \quad \Rightarrow \quad \frac{d\tilde{C}_R}{dt} = -k_1\tilde{C}_R + k_{-1}\tilde{C}_I$$

R.I.P.

$$\Rightarrow \frac{d\tilde{C}_R}{k_1 dt} = -\tilde{C}_R + \frac{k_{-1}}{k_1}\tilde{C}_I$$

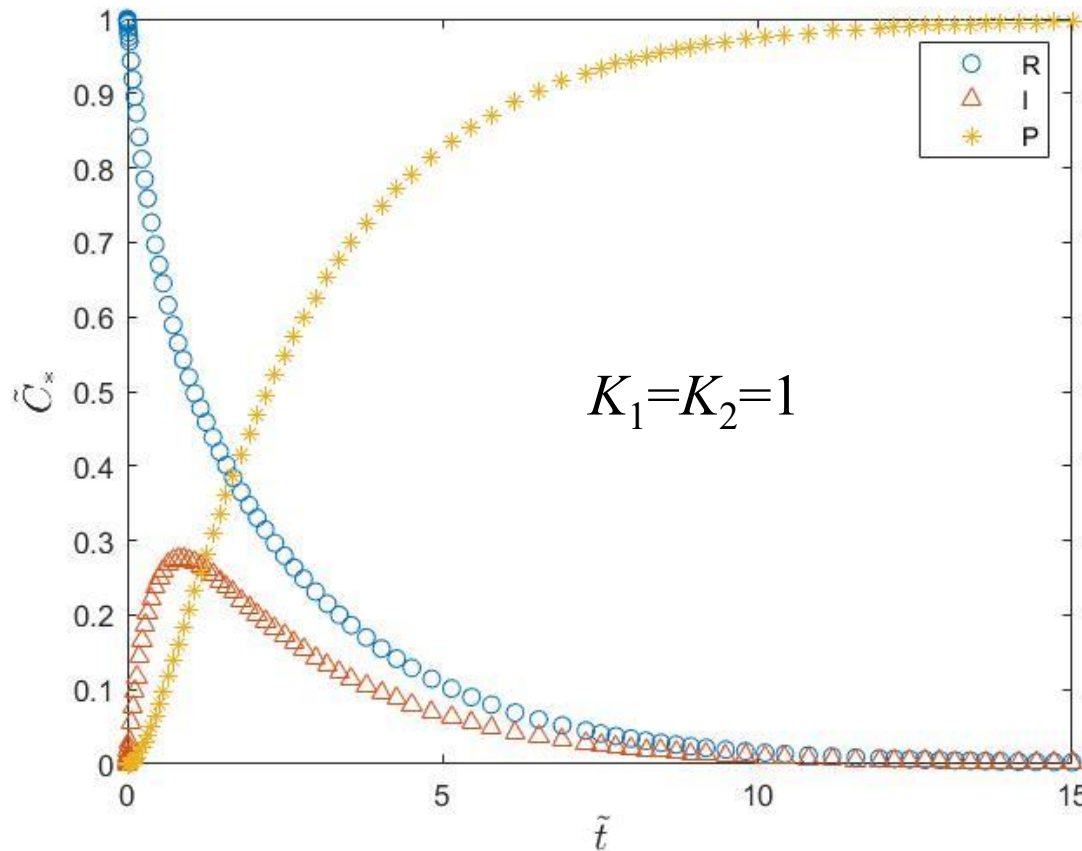
$$\Rightarrow \frac{d\tilde{C}_R}{d\tilde{t}} = -\tilde{C}_R + K_1\tilde{C}_I$$

$$\text{where } \tilde{t} = k_1 t \text{ and } K_1 = \frac{k_{-1}}{k_1}$$

$$\hookrightarrow d(k_1 t) = d\tilde{t}$$

$$\left\{ \begin{array}{l} \frac{dC_R}{dt} = -k_1 C_R + k_{-1} C_I \\ \frac{dC_I}{dt} = k_1 C_R - k_{-1} C_I - k_2 C_I \\ \frac{dC_P}{dt} = k_2 C_I \\ C_R(0) = C_{R,0} \\ C_I(0) = 0 \\ C_P(0) = 0 \end{array} \right. \Rightarrow \left\{ \begin{array}{l} \frac{d\tilde{C}_R}{d\tilde{t}} = -\tilde{C}_R + K_1\tilde{C}_I \\ \frac{d\tilde{C}_I}{d\tilde{t}} = \tilde{C}_R - K_1\tilde{C}_I - K_2\tilde{C}_I \\ \frac{d\tilde{C}_P}{d\tilde{t}} = K_2\tilde{C}_I \text{ for } K_2 = \frac{k_2}{k_1} \\ \tilde{C}_R(0) = 1 \\ \tilde{C}_I(0) = 0 \\ \tilde{C}_P(0) = 0 \end{array} \right.$$

Reaction in Batch Reactor: Plots



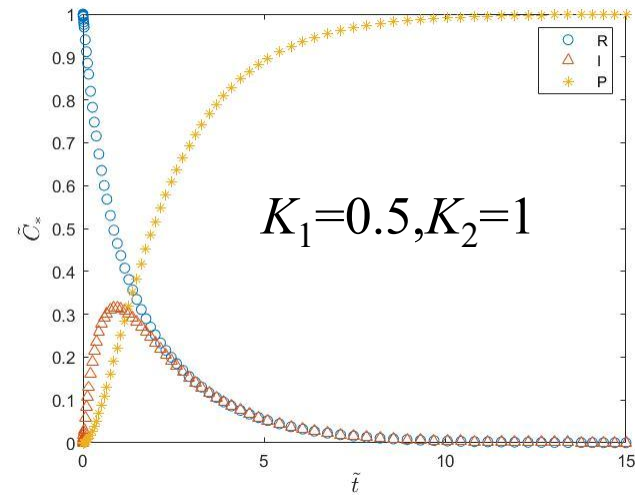
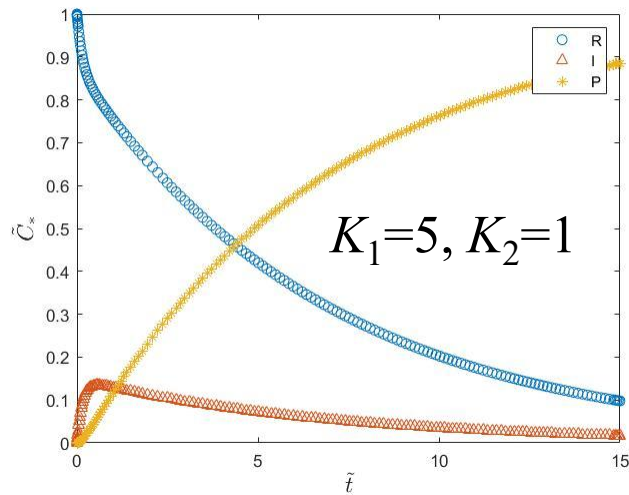
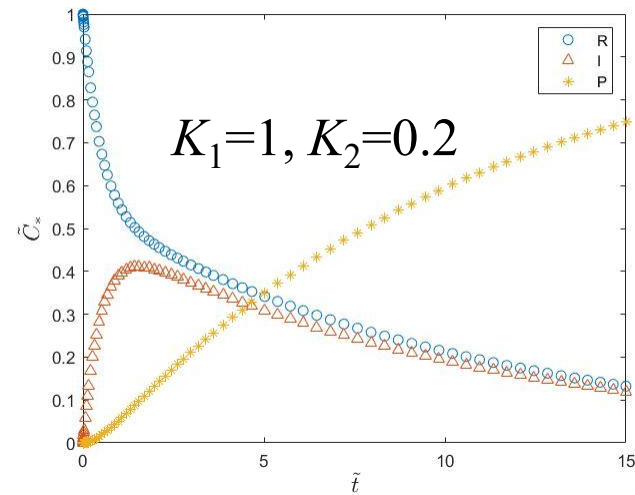
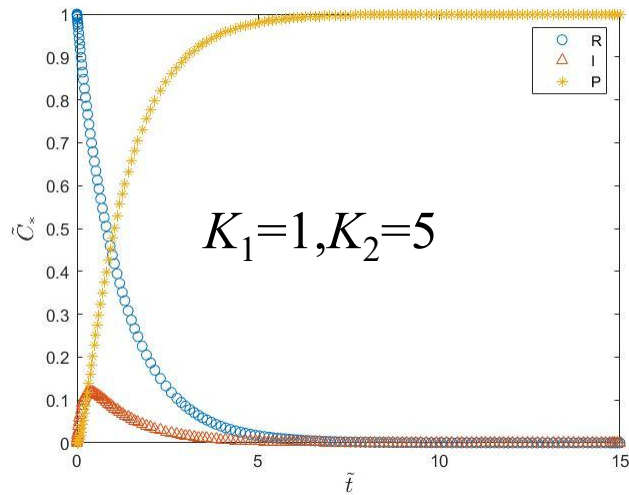
$$K_1 = \frac{k_{-1}}{k_1}$$

$$K_2 = \frac{k_2}{k_1}$$

What happens if K_1 and K_2 vary?

What insights can we derive from the graph?

Reaction in Batch Reactor: Plots



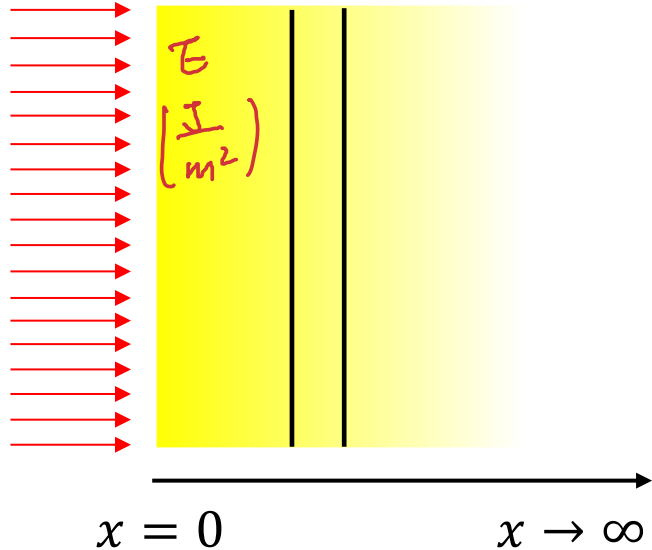
$$K_1 = \frac{k_{-1}}{k_1}$$

$$K_2 = \frac{k_2}{k_1}$$

Non-Steady-State Heat Transfer $\tau(x, t)$

Semi-infinite block 1D case,

Q Energy in. $x \quad x + \Delta x$



Heat eqn.

$$\frac{\partial T}{\partial t} = \kappa \left(\frac{\partial^2 T}{\partial x^2} \right)$$

κ : thermal diffusivity

T : temperature

1. Derive the governing equation for heat transfer (heat equation)
2. Find appropriate BCs and ICs
3. Nondimensionalization

Fourier's Law: $q = -k \frac{dT}{dx}$

q : local heat flux density $\frac{W}{m^2}$

k : material conductivity $\frac{W}{mK}$

$\frac{W}{mK} \cdot \frac{K}{m} = \frac{W}{m^2}$

Derivation: Let's work together!

$$\text{Accumulat}^n = \text{In} - \text{Out} + \text{Generat}^n - \text{Consumpt}^n$$

rate of change of energy

$$-k \frac{\partial T}{\partial x} \bigg|_x \cdot A + \underbrace{\frac{w}{m^2} \cdot m^2 = w = \frac{I}{S}}_{\text{specific heat capacity}} - k \frac{\partial T}{\partial x} \bigg|_{x+\Delta x} \cdot A$$

$$\frac{\partial E}{\partial t} = \frac{\partial (m c T)}{\partial t} = \frac{\partial (P \Delta V c T)}{\partial t} = \frac{\partial (P A \cdot \Delta x c T)}{\partial t} = \underbrace{P \cdot c \cdot A \cdot \Delta x \cdot \frac{\partial T}{\partial t}}_{\text{specific heat capacity}}$$

$$P \cdot c \cdot A \cdot \Delta x \cdot \frac{\partial T}{\partial t} = k \left(\frac{\partial T}{\partial x} \bigg|_{x+\Delta x} - \frac{\partial T}{\partial x} \bigg|_x \right) \cdot A$$

$$\frac{\partial T}{\partial t} = \frac{k}{P \cdot c} \left[\frac{\frac{\partial T}{\partial x} \big|_{x+\Delta x} - \frac{\partial T}{\partial x} \big|_x}{\Delta x} \right] \Rightarrow \frac{\partial^2 T}{\partial x^2} \text{ when } \Delta x \rightarrow 0$$

$$\frac{\partial T}{\partial t} = \left[\frac{k}{P \cdot c} \right] \frac{\partial^2 T}{\partial x^2}$$

k (ka ppa)

ICs: at $t=0$, for any x , $T = T_0$

$$T(t=0) = T_0$$

BCs: ① as $x \rightarrow \infty$, $T = T_0$ or no temp gradient.

$$T(x \rightarrow \infty, t) = T_0$$

② $q|_{x=0} = Q \Rightarrow -k \frac{\partial T}{\partial x} \bigg|_{x=0} = Q$

Nondimensionalization: Let's work together!

$\tilde{T} = \frac{T}{T_0}$, $T = T_0 \tilde{T}$. How to scale x and t ?

$$\frac{\partial T_0 \tilde{T}}{\partial t} = k \frac{\partial^2 (T_0 \tilde{T})}{\partial x^2} \Rightarrow \frac{\partial \tilde{T}}{\partial t} = k \frac{\partial^2 \tilde{T}}{\partial x^2}$$

Look at BC (2) (Neumann)

$$-k \frac{\partial (T_0 \tilde{T})}{\partial x} \Big|_{x=0} = 0 \Rightarrow -\frac{\partial \tilde{T}}{\partial \left(\frac{0x}{kT_0} \right)} \Big|_{x=0} = 1. (*)$$

$$\frac{\partial \tilde{T}}{\partial t} = k \frac{\partial^2 (\tilde{T})}{\partial (L^2 \tilde{x}^2)} \Rightarrow \frac{\partial \tilde{T}}{\partial \left(\frac{k}{L^2} t \right)} = \frac{\partial^2 \tilde{T}}{\partial \tilde{x}^2}$$

$$\tilde{t} = \frac{k}{L^2} t$$

The final dimensionless form:

$$\frac{\partial \tilde{T}}{\partial \tilde{t}} = \frac{\partial^2 \tilde{T}}{\partial \tilde{x}^2}$$

Another way: $\tilde{T} = \frac{T - T_0}{T_0}$ (same dimensionless form for main eqn)

$$\text{but } \tilde{T}(\tilde{x}, 0) = 0$$

$$\tilde{T}(\tilde{x} \rightarrow \infty, \tilde{t}) = 0 \quad \text{or} \quad \frac{\partial \tilde{T}}{\partial \tilde{x}} \Big|_{\tilde{x} \rightarrow \infty} \rightarrow 0$$

$$\frac{\partial^2}{\partial x^2} \Rightarrow \frac{\partial}{\partial x} \left(\frac{\partial}{\partial x} \right)$$

characteristic length
dynamic length.

$$L = \frac{kT_0}{Q}$$

$$\tilde{x} = \frac{0x}{kT_0} = \frac{x}{\frac{kT_0}{Q}} = \frac{x}{L}$$

$$x = L \tilde{x}$$

$$\text{IC1: } T(x, t=0) = T_0 \Rightarrow \frac{T}{T_0}(x, t=0) = 1$$

$$\tilde{T}(\tilde{x}, 0) = 1$$

$$\text{BC1: } T(x \rightarrow \infty, t) = T_0 \Rightarrow \tilde{T}(\tilde{x} \rightarrow \infty, \tilde{t}) = 1$$

$$* \text{ BC2: } -\frac{\partial \tilde{T}}{\partial \tilde{x}} \Big|_{\tilde{x}=0} = 1.$$

Homework Announcement (Week 2)

- **Release Date:** After the lecture on the same day
- **Submission:** Hard copy to be submitted upon arrival to class
- **Due Date:** Week 3 class – 2:00 PM, 11 Feb 2026
- **Grouping:** Clearly indicate all group members' names on the submission.
- **Start Early!** Don't wait until **10 Feb** — trust me, you'll regret it!